How to execute ABM-WP framework step-by-step

This folder contains the files used to carry out the simulations in the article "A Multidimensional Agent-Based Modeling and Simulation Framework for Urban Water Demand Forecast".

Water resource sustainability relies on the interplay of environmental, technological, and social factors shaping urban water demand. However, traditional aggregated water demand forecasting models often fail to capture spatio-temporal heterogeneity and behavioral dynamics of domestic water consumption. We present the Agent-Based Model for Water Prediction (ABM-WP) multidimensional framework, which simulates neighborhood-scale water consumption considering income and population growth with three behavioral profiles: environmentalist, moderate, and wasteful. The ABM-WP implemented on the GAMA platform was tested and calibrated using historical data (2015-2025) from Salvador, Brazil, to generate future scenario analyses for 2026–2035, compared with statistical and machine learning models. ANOVA and Tukey tests establish that the environmentalist scenario (CII) differed significantly from the baseline (p-value = 0.0421) with projected consumption reductions of 48,494 m^3/year. Heatmap analysis revealed the influence of connection density and socioeconomic dynamics on consumption patterns. Results indicate that ABM-WP can contribute to decision support for urban water sustainability.

The main project ABM-WP Github folders and their objectives are described below:\includes\dados\Tabela_consumo_Itapua_120m.csv - Anonimyzed data about individual consumption of households over the last 120 months.

- \includes\dados\Tabela_Subcategoria_Tarifaria.csv – Classification of the tariff category and subcategory types adopted by the local water supply company.
- \includes\dados\Tabela_Coord_Mat.csv - Anonymized geographic coordinates of water connections.
- \includes\dados\Tabela_Consumidores_Itapua_2015_2025_full.csv - Anonymized list of consumers in the Itapuã neighborhood, from 2015 to 2025.
- \includes\Agregados_por_setores_renda_responsavel_BR.csv Income average of responsible of family by Census Sectors 2022 in Brazil.
- \includes\maps\itapua\Itapua.shp - Shapefile with the Census Sectors of the Iapuã Neighborhood.
- \includes\maps\Itapua13.shp - Shapefile with the Census Sectors of the Iapuã Neighborhood
- \includes\maps\LIMITE_BAIRRO.shp - Shapefile of the city of Salvador, with the division of neighborhoods.

You may need to extract the files below, as GitHub does not allow saving them on the server because they exceed the 50MB size limit:

- /includes/ibge_censo2022/Agregados_por_setores_caracteristicas_domicilio1_BR.zip
- /includes/dados/Tabela_Coord_Mat.zip
- /includes/dados/Tabela_consumo_Itapua_120m.zip

Next, the Python script <_Script_prepara_simulacao.ipynb>, located in the Python folder <Python-prepara-simulacao>, must be executed to prepare input files to model execution. <Script_prepara_simulacao.ipynb> will be execute the list of scripts bellow:

Nº	Script	Objetive	Inputs	Outputs
1	Script_gera_consumidores_Itapua_geral.ipynb	Merges raw consumer data with geographic coordinates and tariff metadata using composite keys to prevent row duplication.	includes\dados\Tabela_Consumidores_Itapua_2015_2025_full.csv includes\dados\Tabela_Coord_Mat_Itapua_com_ruido.csv includes\dados\Tabela_Subcategoria_Tarifaria.csv	includes\Tabela_consumidores_Itapua.csv
2	Script_calcula_media_consumo.ipynb	Calculates the average consumption of households, based on 12 months of 2025.	includes\dados\Tabela_consumo_Itapua_120m.csv	includes\Tabela_consumo_medio_Itapua_2025_12m.csv
3	Script_conversor_coordenadas.ipynb	A preprocessing script that utilizes the pyproj library to convert consumer geospatial data from WGS84 (EPSG:4326) to SIRGAS 2000 UTM Zone 24S (EPSG:31984)	includes\Tabela_consumidores_Itapua.csv	includes\Tabela_consumidores_Itapua_convertida .csv
4	Script_mapeia_setor_censitario.ipynb	Identifies the census sector of each consumer unit with a new column CD_SETOR	includes\maps\itapua\Mapa_Itapua.shp includes\Tabela_consumidores_Itapua_convertida.csv	include\Tabela_consumidores_Itapua_com_setor.csv
5	Script_classifica_comportamento_com_media_e_renda_moradores.ipynb	Classifies the behavior and income of each consumer unit, based on average consumption, and adding the columns TP_COMPORTAMENTO and VL_RENDA_MEDIA_RESPONSAVEL.	include\Tabela_consumidores_Itapua_com_setor.csv includes\dados\Tabela_consumo_Itapua_120m.csv includes\ibge_censo2022\Agregados_por_setores_renda_responsavel_BR.csv	includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda.csv includes\dados_gama_detalhados.csv resultados\de_para_setores_consumidores.csv

			includes\ibge_censo2022\Agregados_por_setores_caracteristicas_domicilio1_BR.csv	
6	Script_total_de_ligacoes_por_setor.ipynb	calculates the total number of active connections per sector, generating the output	includes\Tabela_consumidores_Itapua_com_setor.csv resultados\de_para_setores_consumidores.csv	resultados\Tabela_top_10_setores.csv
7	Script_analise_renda_x_consumo_por_setor_censitario.ipynb	Analysis of the correlation between income and consumption, generating scatter plots with an X-axis (average water consumption) and a Y-axis (average income of the head of household).	include\Tabela_consumidores_Itapua_com_setor.csv resultados\de_para_setores_consumidores.csv includes\dados\Tabela_consumo_Itapua_120m.csv include\ibge_censo2022\Agregados_por_setores_renda_responsavel_BR.csv	resultados\resultado_analise_consumo_x_renda_itapua.csv figuras\grafico_resultado_analise_consumo_x_renda_itapua_en.png figuras\grafico_resultado_analise_consumo_x_renda_itapua_pt.png
8	Script_gerar_parametros_renda.ipynb	It generates the income parameters needed for use in simulations, such as lower_income_limit, upper_income_limit, intercept, and gain.	includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda.csv	includes\parametros_renda_calculados_2025.csv includes\parametros_teto_consumo_2025.csv includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2025_ajustado.csv figuras\grafico_analise_parametros.pdf resultados\medias_consumo_por_perfil_2025.csv

After that, open the GAMA platform, import the <ABMS-WP_Framework.gaml> template from the <models> folder and the includes from the <includes> folder, and run the model in GAMA.

Ensure that the variable <data_reference_year> is seted to 2025 to generate future simulations for the next 10 years, i.e., 2026-2035.

Nº	Script	Objetive	Inputs	Outputs
9	models\ABMS-WP_Framework.gaml	Run the scenario simulations using the input data generated in the previous step.	includes\parametros_renda_calculados_2025.csv resultados\medias_consumo_por_perfil_2025.csv includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda.csv includes\Tabela_consumo_medio_Itapua_2025_12m.csv includes\maps\Itapua13.shp includes\maps\LIMITE_BAIRRO.shp	resultados\consumo_previsto_todos_cenarios_2026_2035.csv \resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv

Wait for the simulation to finish running in GAMA, and then open the Python script <_Script_analisa_resultado_simulacao.ipynb>, located in the Python folder <Python-analisa-resultado-simulacao>. It must be executed to generate the graphs and process the results generated by the model. The file <_Script_analisa_resultado_simulacao.ipynb> will execute the scripts listed below:

Nº	Script	Objetive	Inputs	Outputs
10	Script_soma_consumo120m_por_mes.ipynb	This script processes the raw 10-year (120 months) consumption dataset for Itapuã. It aggregates individual meter readings into a consolidated monthly	includes\dados\Tabela_consumo_Itapua_120m.csv	includes\Tabela_consumo_Itapua_120m_por_mes.csv

		time series, summing the total volume consumed per month\year.		
11	Script_comparacao_mensal_consumo_cenarios_lstm_sarima.ipynb	This script generates a comparative line chart of historical versus simulated consumption, sarima, and lstm. Its main feature is no label showed.	resultados\consumo_previsto_todos_cenarios_2026_2035.csv modelos IA\previsoes_futuras_2026_2035.csv modelos IA\previsao_consumo_sarima_2026_2035.csv includes\Tabela_consumo_Itapua_120m_por_mes.csv	figuras\figure-result-simulation-final-comparison-monthly .pdf
12	Script_perfil_consumo_por_setor.ipynb	An analytics script that aggregates classified consumer data by census tracts. It calculates the density of behavioral profiles (Environmentalist, Moderate, Wasteful) for each sector, exports a ranked report of critical areas, and provides global statistics on consumption behavior distribution in Itapuã.	includes\Tabela_consumidores_Itapua_com_setor_e_comportamento.csv	resultados\Resumo_Geral_Comportamento.csv resultados\Relatorio_Comportamento_Por_Setor.csv
13	Script_conexoes_por_bairro.ipynb	Generate char bar of water connections of neighborhood in Salvador	includes\dados\Tabela_ligacoes_agua_Salvador.csv	figuras\figure-neighborhood-distribution-fig15.pdf
14	Script_comparacao_mensal_consumo_cenarios_lstm_sarima_labels.ipynb	This script generates a comparative line chart of simulated consumption, sarima, and lstm. Its main feature is no label showed.	resultados\consumo_previsto_todos_cenarios_2026_2035.csv modelos IA\previsoes_futuras_2026_2035.csv modelos IA\previsao_consumo_sarima_2026_2035.csv	figuras\figure-result-simulation-final-comparison-monthly-labels .pdf
15	Script_projecao_comparativo_diferencas_cenarios_sarima.ipynb	This script produces a PDF chart analyzing the consumption discrepancy between 18 Agent-Based Simulation scenarios and the SARIMA AI forecast from 2026 to 2035. It plots the difference (Scenario_x - SARIMA) for each scenario.	resultados\consumo_previsto_todos_cenarios.csv modelos IA\ previsoes_futuras_2026_2035.csv	figuras\figure-diferenca-cenarios-vs-lstm-fig13.pdf
16	Script_analise_distribuicao.ipynb	This script performs a comprehensive statistical and graphical analysis to evaluate the distribution and normality of the 18 ABM simulation scenarios compared to the model LSTM forecasts.	resultados\consumo_previsto_todos_cenarios_2026_2035.csv modelos IA\ previsoes_futuras_2026_2035.csv	figuras\ normality_analysis_scenarios_vs_lstm.pdf resultados\normality_test_results.csv
17	Script_analise_parametrica_cenarios.ipynb	This script performs a rigorous statistical evaluation to determine if there are significant differences between the 18 Agent-Based Model (ABM) scenarios and the LSTM neural network forecast for the 2026–2035 period.	resultados\consumo_previsto_todos_cenarios_2026_2035.csv modelos IA\previsao_consumo_sarima_2026_2035.csv	Print Table
18	Script_projecao_comparativo_barras_diferencas_cenarios_sarima.ipynb	This script produces a PDF chart bar analyzing the consumption discrepancy between 18 Agent-Based Simulation scenarios and the SARIMA AI forecast from 2026 to 2035. It plots the difference (Scenario_x - SARIMA) for each scenario	resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv modelos IA\previsao_consumo_sarima_2026_2035.csv	figuras\figure-barplot-diferencas-grupos-vs-sarima .pdf

19	Script_boxsplot_comparacao_cenarios.ipynb	This script performs a dual-stage visualization of the water consumption projections (2026–2035) for the 18 ABM scenarios and the LSTM baseline.	resultados\consumo_previsto_todos_cenarios_2026_2035.csv modelos IA\ previsoes_futuras_2026_2035.csv	figuras\figure-boxplot-comparison-scenarios-vs-lstm-fig12.pdf figuras\figure-chartline-comparison-scenarios-vs-lstm-fig12.pdf resultados\resultado_boxsplot_cenarios.csv
20	Script_mapa_calor_impacto_cenarios_horizontal.ipynb	Script to generate heatmap of shift net behavior of profiles after simulations in horizontal chart.	resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv	resultados\figure-net-behavioral-shift-heatmap-h-fig13.pdf resultados\figure-net-behavioral-shift-heatmap-h-fig13.png
21	Script_ Script_projecao_setores_mais_afetados_pos_simulacao.ipynb	Analyzes the changes that have occurred in a given selected sector, generating files with the details of those changes.	resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv	\resultados\analises_mudanca\analise_mudancas_perfil_CVIII.csv \resultados\analises_mudanca\transicoes_detalhadas_CVIII.csv \resultados\analises_mudanca\ matriz_transicao_CVIII.csv \resultados\analises_mudanca\grafico_mudancas_perfil_CVIII.pdf
22	Script_analise_mudanca_perfil.ipynb	Script that generates two reports with ranking of all sector shifts and ranking of sector shifts to wasteful.	resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv	resultados\analises_mudanca\sector_shifts_total_CVIII.csv resultados\ analises_mudanca\sector_shifts_wasteful_CVIII.csv
23	Script_setores_mais_afetados_pos_simulacao_detalhes.ipynb	Analyzes the changes that have occurred in a given selected sector, generating files with the details of those changes.	resultados\comparativo_perfil_cenarios_qgis_2026_2035.csv includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2025_ajustado.csv	resultados\ analises_mudanca\matriculas_mudancas_{setor}_{cenario}_{timestamp}.csv resultados\ analises_mudanca\matriculas_mudancas_{setor}_{cenario}_resumo_{timestamp}.csv resultados\ analises_mudanca\dados_cruzados_mudancas_{setor}_{cenario}_{timestamp}.csv

To project the simulation of the actual consumption period from 2016 to 2025, follow the steps below:

the Python script <_Script_prepara_simulacao.ipynb>, located in the Python folder <Python-prepara-simulacao-2015>, must be executed to prepare input files to model execution. <Script_prepara_simulacao.ipynb> will be execute the list of scripts bellow:

Nº	Script	Objetive	Inputs	Outputs
1	Script_gera_consumidores_Itapua_geral.ipynb	Merges raw consumer data with geographic coordinates and tariff metadata using composite keys to prevent row duplication.	includes\dados\Tabela_Consumidores_Itapua_2015_2025_full.csv includes\ dados\Tabela_Coord_Mat_Itapua_com_ruido.csv includes\ dados\Tabela_Subcategoria_Tarifaria.csv	includes\Tabela_consumidores_Itapua.csv
2	Script_calcula_media_consumo_2015.ipynb	Calculates the base consumption average for the year 2015.	includes\dados\Tabela_consumo_Itapua_120m.csv	includes\Tabela_consumo_medio_Itapua_2015.csv
3	Script_gera_consumidores_Itapua_2015.ipynb	Consolidates the master table with UTM coordinates, census tracts (IBGE), and behavioral imputation for the	includes\Tabela_consumo_medio_Itapua_2015_12m.csv	includes\ Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015.csv

		2015 baseline. Convert consumer geospatial data from WGS84 (EPSG:4326) to SIRGAS 2000 UTM Zone 24S (EPSG:31984)	includes\Tabela_consumidores_Itapua.csv includes\censo_ibge2022\Agregados_por_setores_renda_responsavel_BR.csv includes\censo_ibge2022\Agregados_por_setores_caracteristicas_domicilio1_BR .csv Includes\maps\itapua\Mapa_Itapua.shp	
4	Script_analise_renda_x_consumo_por_setor_censitario.ipynb	Analysis of the correlation between income and consumption, generating scatter plots in English with an X-axis (average water consumption) and a Y-axis (average income of the head of household).	include\Tabela_consumidores_Itapua_com_setor.csv includes\Tabela_consumo_Itapua_60m.csv include\ibge_censo2022\Agregados_por_setores_renda_responsavel_BR.csv	resultados\resultado_analise_consumo_x_renda_itapua.csv figuras\grafico_ resultado_analise_consumo_x_renda_itapua_en.png figuras\grafico_ resultado_analise_consumo_x_renda_itapua_pt.png
5	Script_gerar_parametros_renda_2015.ipynb	It generates the income parameters needed for use in simulations, such as lower_income_limit, upper_income_limit, intercept, and gain.	includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015.csv includes\Tabela_consumo_medio_Itapua_2015_12m.csv	includes\parametros_renda_calculados_2015.csv figuras\grafico_analise_renda_2015.png resultados\medias_consumo_por_perfil_2015.csv includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015_ajustado.csv includes\parametros_teto_consumo_2015.csv

After that, open the GAMA platform, import the <ABMS-WP_Framework.gaml> template from the <models> folder and the includes from the <includes> folder, and run the model in GAMA.

Ensure that the variable <data_reference_year> is seted to 2015 to generate past simulations for the next 10 years, i.e., 2016-2025.

Nº	Script	Objetive	Inputs	Outputs
6	models\ABMS-WP_Framework.gaml	This script processes the raw 10-year (120 months) consumption dataset for Itapuã. It aggregates individual meter readings	includes\parametros_renda_calculados_2015.csv resultados\medias_consumo_por_perfil_2015.csv includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015_ajustado.csv includes\Tabela_consumo_medio_Itapua_2015.csv includes\maps\Itapua13.shp includes\maps\LIMITE_BAIRRO.shp	resultados\simulacao_hindcasting_2016_2025.csv resultados\comparativo_perfil_cenarios_qgis_2016_2025.csv

Wait for the simulation to finish running in GAMA, and then open the Python script <_Script_analisa_resultado_simulacao.ipynb>, located in the Python folder <Python-analisa-resultado-simulacao-2015>. It must be executed to generate the graphs and process the results generated by the model. The file <_Script_analisa_resultado_simulacao.ipynb> will execute the scripts listed below:

Nº	Script	Objetive	Inputs	Outputs
7	Script_analise_renda_x_consumo_por_setor_censitario_2015.ipynb	Analysis of the correlation between income and consumption, generating scatter plots in Portuguese	includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015.csv	resultados\resultado_analise_consumo_x_renda_itapua.csv

		and English with an X-axis (average water consumption) and a Y-axis (average income of the head of household).		figuras\grafico_ resultado_analise_consumo_x_renda_itapua_en.pdf
8	Script_gera_consumo_real_historico.ipynb	Aggregates actual historical water consumption from 2016 to 2025. Creating a historical baseline (Ground Truth) to validate hindcasting simulation results.	includes\Tabela_consumidores_Itapua_com_setor_comportamento_e_renda_2015.csv includes\dados\Tabela_consumo_Itapua_120m.csv	includes\Tabela_consumo_real_historico_2016_2025.csv
9	Script_compara_resultados_transicao_perfil_2016_2025_3_graficos_mensal_simbolos.ipynb	Generates a retrospective forecast validation chart from 2016 to 2025 with the simulation results and the historical consumption for profile (Enviromentalist, Moderate, Wasteful).	includes\Tabela_consumo_real_historico_2016_2024 .csv resultados\simulacao_hindcasting_2016_2024.csv	figuras\figure-validation-profile-monthly-PERDULARIO.pdf figuras\figure-validation-profile-monthly-AMBIENTALISTA.pdf figuras\figure-validation-profile-monthly-MODERADO.pdf
10	Script_compara_resultados_mensal_2016_2025_lstm_sarima_sem_os_cenarios.ipynb	Generates a graph showing the consumption from 2016 to 2025, compared to LSTM and SARIMA models.	includes\Tabela_consumo_real_historico_2016_2025.csv resultados\simulacao_hindcasting_2016_2025.csv includes\previsao_consumo_sarima_2016_2025.csv includes\dados\previsao_consumo_2016_2025.csv	figuras\figure-hindcasting-validation-monthly-with-lstm-sarima.pdf